

Matthew Topham

AI Systems & Full-Stack Developer

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Computer science student at CU Boulder building AI tools for spatial creation, wearable context, and interactive storytelling. Work spans TypeScript interfaces, Python services, model integration, persistent state, and automated validation.

SELECTED PROJECTS

Matrix Loading Operator · AI spatial creation

Creator · 2026 · v1.1 released September 2026

- Built a persistent Three.js/WebXR world for desktop, VR, and AR, connecting voice and text requests to Codex, procedural tools, and Blender asset workflows.
- Implemented active-turn steering, guarded placement on measured support surfaces, and room-relative AR layout controls with save/reopen persistence.
- Shipped v1.1 with 750 client, 897 service, and 11 release-builder tests passing; verified reproducible packages and an extracted-service smoke test.

Demo · v1.1 release and validation · v1.2 in progress

Manfred · Wearable AI memory and context

Creator · 2026 · Prototype

- Built an Android/Flutter companion and Python services for wearable audio ingestion, transcription, searchable context, and durable upload buffering.
- Connected E09 smart glasses to ChatGPT Live through phone automation; voice works, while photo capture and delivery remain under refinement.

OPEN-SOURCE EXPERIENCE

Marinara Engine + Agents · Pasta Devs

Open-Source Developer · 2026–Present

- Contributed 100 merged pull requests across Engine (68) and Agents (32), spanning React/TypeScript UI, Node.js services, persistent world state, and regression tests.
- Built hierarchical World Maps and versioned host APIs for downloadable agents, including persistence, transaction handling, and recovery paths.
- Implemented image-aware storyboard planning that grounds animation in generated keyframes; packaged live-tested MiniMax H3 prompts as reusable Storyboard choices.

Selected work: World Maps · host APIs · animation planning · H3 prompts

ADDITIONAL WORK

Demerzel: Built conversation-to-wiki ingestion, daily AI briefs, service health checks, and GPU workload controls for a local AMD/NVIDIA compute fleet.

School of the Ancients: Built web and VR learning prototypes with AI historical mentors using React, Node.js, Supabase/Postgres, real-time voice AI, and Unity.

TECHNICAL SKILLS

Software: TypeScript, JavaScript, Python, SQL, React, Node.js, Flutter, REST, WebSockets.

Spatial / AI / infrastructure: Three.js, WebXR, Blender, local inference, ComfyUI, Linux, Docker, GitHub Actions, Playwright, regression testing.

EDUCATION & CERTIFICATION

University of Colorado Boulder — B.A. Computer Science, in progress

Front Range Community College — A.S. Computer Science, 2025

NVIDIA-Certified Associate — AI Infrastructure & Operations, August 2025